

CL : PYTHON PROGRAMMING LABORATORY (MED23CL202)

COURSE OUTCOME	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
Create manipulate and use various data types efficiently and develop modular, reusable and efficient code using functions	H											
Design the program models using object-oriented programming concepts such as encapsulation, inheritance and polymorphism as used in python			H									
Explore different file operation		M										
Implement error handling in python		L										
Understand the importance of data analysis and implementation of different visual components			M									

Second Year - Semester III

C. No.	Course Code	Course category	Course Title	L	T	P / RP	Total Credits	Total Hours/ Marks
36	MED23CL202	CL	Python Programming Laboratory	0	0	1	1	30 / 100

Learning Outcomes

At the end of the course, the student should be able to

CO1: Create manipulate and use various data types efficiently and develop modular, reusable and efficient code using functions

CO2: Design the program models using Object-Oriented Programming concepts such as encapsulation, inheritance and polymorphism as used in Python

CO3: Explore different file operation

CO4: Implement error handling in Python

CO5: Understand the importance of data analysis and implementation of different visual components

List of Experiments

1 Fundamentals of Python (6 Hours)

Data Types – List, Tuple, Dictionaries and set - Loops - Conditional statement - String Manipulation - Function and Module

2 OOPs Paradigm (6 Hours)

Class - Object - Data Abstraction - Data Encapsulation - Inheritance - Polymorphism

3 File Handling (6 Hours)

Reading a file - Writing a file - File operations

4 Exception Handling (6 Hours)

Exception - Types of Exception - Exception Handling - User-defined Exception

5 Data Analysis and Visualization (6 Hours)

Pandas - Matplotlib – Basic Charts - Seaborn – Distribution Plots - Categorical plots - Interactive Visualization

REFERENCES:

1.	Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", Green Tea Press, 2 nd Edition, 2015.
2.	Charles Dierbach, "Introduction to Computer Science Using Python", Wiley India Pvt Ltd, 1 st Edition, 2015.
3.	Wesley J Chun, "Core Python Applications Programming", Pearson Education, 3 rd Edition, India, 2015.